# Student Course Registration — Test Report

Scope: black-box and white-box pytest coverage for registration.py.

```
Initial black-box run (35 tests): 27 passed, 8 failed.
  Statements: 54/57 (94.7%)    Branch outcomes: 23/26 (88.5%)
 Combined pytest-cov coverage: 93%

Full suite on the github-actions branch (43 tests): 33 passed, 10 failed.
  Statements: 55/57 (96.5%)    Branch outcomes: 24/26 (92.3%)
 Combined pytest-cov coverage: 95%
```

The supplied implementation intentionally contains defects. Failing tests are preserved as evidence; expected outputs were not altered to make them pass.

Detailed command output is saved in coverage_initial.txt and coverage_final.txt.

# Coverage gaps and specification mismatches

`Initial black-box run uncovered registration.py lines 51, 59, and 71.`

`Full suite uncovered lines 59 and 71.`

Line 51 — DUES_BLOCK return:

The implementation checks dues > 1000; FR-04 requires blocking any positive dues balance.

Line 59 — WAITLISTED return:

The implementation checks enrolled > capacity. Valid full courses have enrolled == capacity; over-capacity inputs are rejected by validation first. This makes the decision path unreachable for valid inputs.

Line 71 — waitlist append:

The append path cannot be reached for a valid full course while the decision function does not return WAITLISTED at capacity.

The separate whitebox-testing branch adds isolated structural tests for both waitlist update outcomes.

# Reported failing test cases

A2-TID-2 — FR-01: an allowed total of 18 credits returns False.

A2-TID-8 / DT-01 / DT-07 / DT-08 — FR-04: positive dues do not block first.

A2-TID-9 / DT-05 — FR-08: a full course returns REGISTERED, not WAITLISTED.

A4 register waitlist — FR-12: waitlist state is not updated for full courses.

A4 failed registration (DUES_BLOCK) — FR-12: registration state changes.

A4 withdraw enrolled — FR-13: course enrollment is not decremented.

WB-06 — waitlist duplicate test cannot get WAITLISTED from current decision.

WB-08 — FR-03: partially completed prerequisites are accepted.

Refer to the raw reports and GitHub defect Issues for exact tracebacks and reproduction steps.